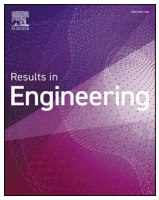
Results in Engineering 29 (2026) 109096



Contents lists available at ScienceDirect

Results in Engineering

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A refined semi-analytical model for hydraulic analysis of laminar power-law fluid flow in eccentric annuli

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ARTICLE INFO

Keywords:
Eccentric annulus
Power-law fluid
Laminar flow
Hydraulic parameters
Superposition of concentric annuli

ABSTRACT

Predicting laminar power-law fluid flow in eccentric annuli is crucial for managing wellbore pressure in complex drilling operations. This study presents a refined semi-analytical model that rigorously implements the concentric annular superposition principle. The core novelty lies in a hybrid numerical-analytical technique that employs hypergeometric functions and the bisection method to precisely locate the zero-shear-stress position within each discretized element, thereby overcoming the inaccuracies of previous approximation-based methods. This framework enables the direct and accurate calculation of key hydraulic parameters—including asymmetric velocity profiles, shear distributions, and frictional pressure drops—from given engineering flow rates. The model's high precision is demonstrated by reducing velocity continuity errors at the zero-shear interface to below 0.0007 m/s. Key findings confirm that peak velocity and shear occur on the wide-gap side, while increasing eccentricity significantly reduces frictional pressure gradients. Notably, at high eccentricities ($\varepsilon > 0.4$), the influence of the flow behavior index (n) becomes more pronounced, with fluids exhibiting higher n values showing lower frictional pressure gradients.

1. Introduction

Understanding key hydraulic parameters, such as shear stress, shear rate, velocity, and frictional pressure drop gradient, for power-law drilling fluids in eccentric annuli is the theoretical foundation for accurate ECD prediction, wellbore pressure management, and effective hole cleaning in complex trajectory wells, including extended-reach and horizontal wells [1,2]. Effective wellbore pressure management prevents drilling complications, including kicks and lost circulation, ensuring operational safety and efficiency [3]. Extensive theoretical studies, numerical simulations, and experimental results have demonstrated that both the pressure loss and velocity distribution of annular flow are not only dependent on the fluid's rheological parameters(consistency index and flow behavior index), but are also significantly affected by annular geometry, particularly the inner to outer diameter ratio, and drill string eccentricity [4,5].

The primary methodologies for investigating eccentric annular flow can be broadly categorized into three groups: (i) numerical simulation models, (ii) semi-analytical models, and (iii) empirical or semi-empirical correlations.

Numerical simulation, employing techniques such as the finite difference method with bipolar or boundary-fitted coordinates [6–9] or Computational Fluid Dynamics (CFD) [5,10–12], represents the most robust approach. These models can handle complex rheologies, such as the Yield-Power-Law (YPL) model [9,13], and intricate geometries, including partially blocked annuli [8]. They provide high-fidelity resolution of the entire flow field. However, their primary drawback is the significant computational expense and time-intensive nature, which currently limits their direct deployment for real-time hydraulic analysis in operational scenarios.

At the other end of the spectrum, empirical correlations offer rapid calculations for engineering practice. Classic works by Haciislamoglu and Cartalos [14] and more recent developments by Dokhani et al. [9] provide practical equations to estimate the frictional pressure drop. While computationally efficient, these correlations are essentially "black boxes" derived from regression analysis of experimental or numerical data. They do not provide detailed physical insights into the flow field, such as velocity and shear rate profiles, which are critical for applications like cuttings transport and barite sag analysis.

Semi-analytical models, such as the variable-height narrow slot

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https://doi.org/10.1016/j.rineng.2026.109096
Received 9 July 2025; Received in revised form 1 January 2026; Accepted 12 January 2026
Available online 12 January 2026
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approximation [15–19] and the concentric annular superposition method [20,21], occupy a middle ground. The narrow slot model, although simple, yields unrealistic, symmetric velocity profiles and linearized shear stress distributions[22]. In contrast, the concentric annular superposition method, pioneered by Luo and Peden [21], discretizes the eccentric annulus into a series of concentric annular elements. This approach can capture the asymmetric nature of the flow, offering a better balance between computational complexity and physical accuracy. However, a significant challenge in applying the superposition method lies in accurately determining the location of zero shear stress within each element. Previous implementations, including the work by Luo and Peden [21], often employed explicit approximation formulas for this position to simplify the complexity of solving an implicit integral equation. This approximation, while simplifying computation, introduces errors in the velocity profile and, consequently, the final pressure drop calculation. Furthermore, many existing superposition-based models focus on calculating dimensionless flow parameters under an assumed pressure gradient, lacking a direct methodology to compute the frictional pressure drop for a given volumetric flow rate—a critical requirement for engineering design.

This study addresses this gap by proposing a refined semi-analytical model for laminar power-law fluid flow. While acknowledging that advanced numerical models have addressed more complex YPL fluids [9], the power-law model remains fundamental and widely used. The core novelty of this work lies not in the superposition concept itself, but in its rigorous implementation. We introduce a hybrid numerical-analytical technique that uses hypergeometric functions to express the velocity integral analytically and employs the bisection method to numerically pinpoint the exact location of zero shear stress within each element. This avoids the inaccuracies of previous approximation methods. The developed framework enables the direct and efficient calculation of detailed hydraulic profiles and frictional pressure drops from operational inputs (i.e., flow rates), thus bridging the gap between computationally expensive numerical simulations and oversimplified empirical correlations.

2. Method of concentric annuli superposition

The axial velocity of the drilling fluid in eccentric annuli depends on both the radial position, r, and the angular coordinate, θ, making direct analytical solutions to the momentum equations mathematically intractable. To address this challenge, the eccentric annular geometry is approximated as a series of quasi-concentric annular subregions, effectively decomposing the complex eccentric flow problem into multiple tractable subproblems that can be solved through existing concentric annular solutions. The total volumetric flow rate is determined by calculating the flow rate within each discretized subregion and performing algebraic summation. This computational framework establishes the functional relationship between the frictional pressure drop gradient of the eccentric annuli and the global volumetric flow rate.

2.1. Spatial domain decomposition

The geometric configuration is defined with the drill string radius r_1, wellbore radius r_2, and drill string standoff e. A polar coordinate system is established with its origin O at the drill string center, where the circumferential angle θ equals 0, corresponding to the high-side direction. This angle increases counterclockwise within the eccentric annular cross-section, as illustrated in Fig. 1. The annular region is partitioned into infinitesimal quadrilateral elements through an infinite set of radial lines originating from O, bounded by inner and outer circumferential arcs. Each quadrilateral element comprises an inner arc, an outer arc, and two radial boundary lines. For a radial line OA_i, an arbitrary angle θ_i, its intersections with the inner and outer walls are denoted as C_i and A_i, respectively,defining the local annular clearance h_i.Similarly, the adjacent radial line OA_{i-1} at angle θ_{i-1} intersects the boundaries at C_{i-1} and A_{i-1}. The resultant trapezoidal segment(highlighted in red) represents a characteristic differential element for hydraulic analysis.

The radial coordinate extending along angular position θ_i from the origin O to the outer boundary is defined as $r_{2e,i}$:

$$r_{2e,i} = r_{2e}(\theta_i) = e\cos\theta_i + \sqrt{r_2^2 - e^2\sin^2\theta_i} \tag{1}$$

The annular clearance h_i at angle θ_i is:

$$h_i = r_{2e,i} - r_1 \tag{2}$$

2.2. Solution strategy

Subdivide the eccentric annulus circumferentially into infinitesimal

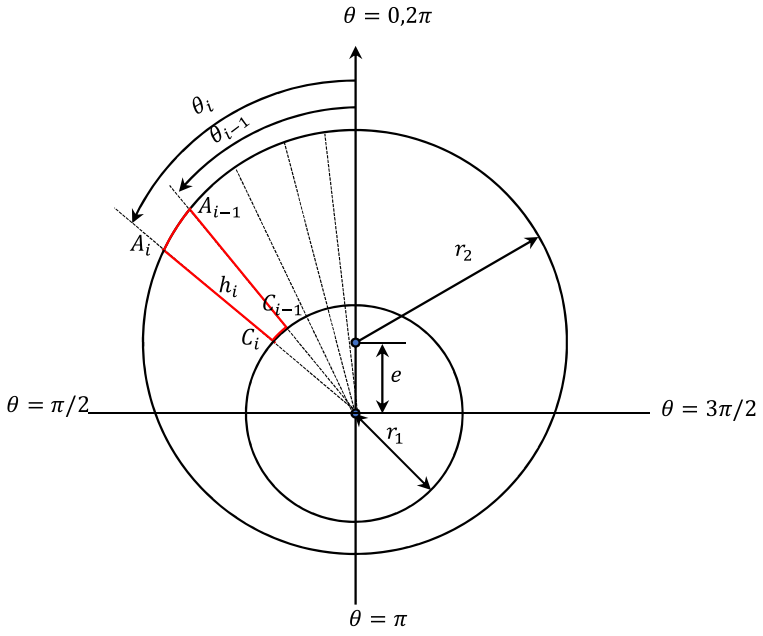


Fig. 1. Cross-section of eccentric annulus.

elements. For each elemental segment at an angular position θ_i, the flow problem is approximated as that within an equivalent concentric annulus, defined by a constant inner radius r_1 and an angularly dependent outer radius $r_{2e,i} = r_1 + h_i$. Upon this geometric simplification, the original governing equations for the eccentric annulus, characterized by coupled r and θ dependencies, reduce to purely r dependent equations applicable to concentric annuli. Consequently, solving each equivalent concentric annulus problem using its relatively simple equations and then integrating the local solutions over the circumferential domain, the global flow behavior of the eccentric annulus is reconstructed.

3. Velocity profile

The constitutie equation for power-law fluids is expressed as:

$$\tau = -k\left|\frac{dv}{dr}\right|^{n-1}\frac{dv}{dr}, n \le 1 \tag{3}$$

Where τ is the shear stress(Pa), k is the consistency index(Pa·s^n), dv/dr is the shear rate (s^{-1}).

For the element at circumferential angle θ_i in the eccentric annulus, the radial position of the zero shear stress point is denoted as $r_{oe,i}$. Based on the classic solution for power-law fluid flow in a concentric annulus by Hanks and Larsen [23], and applying it to an individual element within the superposition framework as demonstrated by Luo and Peden [21], the velocity expressions for the two intervals within the sub-element, bounded by $r_{oe,i}$ are obtained as follows:

$$v_{e,i}(r_e) = \left(\frac{G_{pfe}}{2k}\right)^{1/n}\int_{r_1}^{r_e}\left[\frac{(r_{oe,i})^2}{r_e} - r_e\right]^{1/n} dr_e \left(for\ r_1 \le r_e \le r_{oe,i}\right) \tag{4a}$$

By introducing the variable substitutions $s = \frac{1}{n}, t = \frac{r_e}{r_{0e,i}}$,the integral transforms to:

$$I_i = r_{oe}^{S+1}\int_{r_1/r_{oe,i}}^{r_e/r_{oe,i}} \frac{(1-t^2)^s}{t^s}dt$$

Let $u = t^2$,the integral becomes:

$$I_i = \frac{1}{2}r_{oe}^{s+1}\int_{(r_1/r_{oe,i})^2}^{(r_e/r_{oe,i})^2} u^{-\frac{s+1}{2}}(1-u)^s du$$

Applying the linearity of integration and splitting the limits:

$$I_i = \frac{1}{2}r_{oe}^{s+1}\left[\int_0^{(r_e/r_{oe})^2} u^{-\frac{s+1}{2}}(1-u)^s du - \int_0^{(r_1/r_{oe})^2} u^{-\frac{s+1}{2}}(1-u)^s du\right]$$

The integral form mathes the incomplete beta function [24],allowing the velocity bo be expressed as:

$$v_{e,i} = \frac{r_{oe,i}^{s+1}}{2}\left[B\left(r_e^2 \Big/ r_{oe,i}^2, \frac{n-1}{2n}, \frac{n+1}{n}\right) - B\left(r_1^2 \Big/ r_{oe,i}^2, \frac{n-1}{2n}, \frac{n+1}{n}\right)\right]$$

For drilling fluids,$n < 1$,(i.e.,$a < 0$), and the integrand diverges. Thus, it is further transformed into a hypergeometric function [24].

Given the identity $B(x,a,b) = \frac{x^a}{a}{}_2F_1(a,1-b;a+1;x)$,and substituing s with n:

$$I_i = \frac{n r_{oe}^{\frac{2}{n}}}{n-1}\left[(r_e)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n}, -\frac{1}{n}; \frac{3n-1}{2n}; \frac{r_e^2}{r_{oe}^2}\right) - (r_1)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n}, -\frac{1}{n}; \frac{3n-1}{2n}; \frac{r_1^2}{r_{oe}^2}\right)\right]$$

$$v_{e,i}(r_e) = \left(\frac{G_{pfe}}{2k}\right)^{1/n}\int_{r_e}^{r_{2e,i}}\left[r_e - \frac{(r_{oe,i})^2}{r_e}\right]^{1/n} dr_e \ \left(for\ r_{oe,i} \le r_e \le r_{2e,i}\right) \tag{4b}$$

Where r_e denotes the radial coordinate within the sub element(m),$v_{e,i}$ represents the velocity at angle θ_i(m/s), and G_{pfe} is the frictional pressure gradient for eccentric flow(Pa/m).

Substituting expression of I_i into expression (4a), yields:

$$v_{e,i} = \left(\frac{Gp_f}{2k}\right)^{1/n}\frac{n r_{oe,i}^{\frac{2}{n}}}{n-1}\left[(r_e)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n}, -\frac{1}{n}; \frac{3n-1}{2n}; \frac{r_e^2}{r_{oe,i}^2}\right) - (r_1)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n}, -\frac{1}{n}; \frac{3n-1}{2n}; \frac{r_1^2}{r_{oe,i}^2}\right)\right] \tag{5a}$$

This paper derives the hypergeometric function expressions for Eqs. (4a) and (b) due to the difficulty in obtaining analytical integral solutions.

3.1. Velocity in the interval $r_1 \le r_e \le r_{oe,i}$

The integral in Eq. (4a) is defined as:

$$I_i = \int_{r_1}^{r_e}\left[\frac{(r_{oe,i})^2}{r_e} - r_e\right]^{1/n} dr_e$$

Note that this expression is only applicable when$(3n-1)/(2n) > 0$,i.e., $n > 1/3$. For drilling fluids, the flow behavior index n typically exceeds 0.52, ensuring the engineering applicability of this formulation.

The hypergeometric function can be computed by the following formula:

$${}_2F_1(a,b;c;z) = \sum_{k=0}^{\infty}\frac{(a)_k(b)_k}{(c)_k}\frac{z^k}{k!} \tag{6}$$

Where $(a)_k$ denotes the Pochhammer symbol (rising factorial).

3.2. Velocity in the interval $r_{oe,i} \le r_e \le r_{2e,i}$

The integral in Eq. (4a) is defined as:

$$I_o = \int_{r_e}^{r_{2e,i}} \left[r_e - \frac{\left(r_{oe,i}\right)^2}{r_e} \right]^{1/n} dr_e$$

Let $t = \frac{r_e}{r_{oe,i}}$,the integral transforms to:

$$I_o = r_{oe,i}^{1+\frac{1}{n}} \int_a^b \left(t - \frac{1}{t} \right)^{1/n} dt, a = \frac{r_e}{r_{oe,i}}, b = \frac{r_{2e}}{r_{oe,i}}$$

Let $y = t^2$ and perform the integral transformation:

$$I_o = \frac{1}{2} r_{oe,i}^{\frac{n+1}{n}} \int_{r_e^2/(r_{oe})^2}^{(r_{2e})^2/(r_{oe})^2} (y-1)^{1/n} y^{-\frac{n+1}{2n}} dy$$

$$I_o = \frac{n}{2(n+1)} \left(r_{oe,i}\right)^{\frac{n+1}{n}} \left[\left(\frac{r_{2e,i}^2 - \left(r_{oe,i}\right)^2}{r_{2e,i}^2} \right)^{\frac{n+1}{n}} {}_2F_1\left(\frac{n+1}{n}, \frac{3n+1}{2n}; \frac{2n+1}{n}; \frac{r_{2e,i}^2 - \left(r_{oe,i}\right)^2}{r_{2e,i}^2} \right) - \left(\frac{r_e^2 - \left(r_{oe,i}\right)^2}{r_e^2} \right)^{\frac{n+1}{n}} {}_2F_1\left(\frac{n+1}{n}, \frac{3n+1}{2n}; \frac{2n+1}{n}; \frac{r_e^2 - \left(r_{oe,i}\right)^2}{r_e^2} \right) \right]$$

Introduce the variable substitution $z = \frac{y-1}{y}$, which yields $y = \frac{1}{1-z}$, and $dy = \frac{dz}{(1-z)^2}$. Substituting these into the integral and simplifying, we obtain:

$$I_o = \frac{1}{2} \left(r_{oe}\right)^{\frac{n+1}{n}} \int_{z_1}^{z_2} z^{\frac{1}{n}} (1-z)^{-\frac{3n+1}{2n}} dz$$

where the integration limits are: $z_1 = 1 - \frac{r_{oe,i}^2}{r_e^2}, z_2 = 1 - \frac{r_{oe}^2}{(r_{2e})^2}$.

Using the definition of the incomplete beta function:

$$B(z; a, b) = \int_0^z t^{a-1} (1-t)^{b-1} dt$$

Compare the integrand $z^{\frac{1}{n}} (1-z)^{-\frac{3n+1}{2n}}$ to identify the parameters:

$$a = \frac{1}{n} + 1, b = -\frac{n+1}{2n}.$$

Since $b < 0$, reformulate the integral in terms of the hypergeometric function:

$$\int_{z_1}^{z_2} z^{\frac{1}{n}} (1-z)^{-\frac{3n+1}{2n}} dz = \frac{z_2^{\frac{1}{n}+1}}{\frac{1}{n}+1} \cdot {}_2F_1\left(\frac{1}{n}+1, \frac{3n+1}{2n}; \frac{1}{n}+2; z_2 \right) - \frac{z_1^{\frac{1}{n}+1}}{\frac{1}{n}+1} \cdot {}_2F_1\left(\frac{1}{n}+1, \frac{3n+1}{2n}; \frac{1}{n}+2; z_1 \right)$$

Substituting this result back into the original integral expression and simplifying:

Substituting it back into Eq. (4b) yields:

$$v_{e,i} = \left(\frac{Gp_f}{2k} \right)^{1/n} \frac{n}{2(n+1)} \left(r_{oe,i}\right)^{\frac{n+1}{n}} \left[\left(\frac{r_{2e,i}^2 - \left(r_{oe,i}\right)^2}{r_{2e,i}^2} \right)^{\frac{n+1}{n}} {}_2F_1\left(\frac{n+1}{n}, \frac{3n+1}{2n}; \frac{2n+1}{n}; \frac{r_{2e,i}^2 - \left(r_{oe,i}\right)^2}{r_{2e,i}^2} \right) - \left(\frac{r_e^2 - \left(r_{oe,i}\right)^2}{r_e^2} \right)^{\frac{n+1}{n}} {}_2F_1\left(\frac{n+1}{n}, \frac{3n+1}{2n}; \frac{2n+1}{n}; \frac{r_e^2 - \left(r_{oe,i}\right)^2}{r_e^2} \right) \right] \quad (5b)$$

4. Numerical computation of zero shear stress position

Replacing r_2 in the formula for the zero shear stress position in a concentric annulus [23] r_o with $r_{2e,i}$, the $r_{oe,i}$ of the sub-element i., in an eccentric annulus, can be estimated with Eq. 7 approximately. However, based on this estimated value, the velocities calculated by Eqs. (5a) and (b) at $r_{oe,i}$ are unequal, failing to meet the physical condition of equal flow velocities on both sides of the zero shear stress point. This discrepancy introduces errors in the calculated results for frictional pressure drop gradient, velocity profile, shear stress distribution, and shear rate distribution, derived from the estimated value.

$$r_{oe,i} = \left(\frac{r_{2e,i}^2 - r_1^2}{2ln\frac{r_{2e,i}}{r_1}} \right)^{\frac{1}{2}} \quad (7)$$

At $r_e = r_{oe,i}$,by equating the velocities from expressions (5a) and (5b),

the exact position of $r_{oe,i}$ can be determined, yielding the following equation:

$$\frac{nr_{oe,i}^{\frac{2}{n}}}{n-1}\left[\left(r_{oe,i}\right)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n},-\frac{1}{n};\frac{3n-1}{2n};1\right)-\left(r_1\right)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n},-\frac{1}{n};\frac{3n-1}{2n};\frac{r_1^2}{r_{oe,i}^2}\right)\right]$$
$$=\frac{n}{2(n+1)}\left(r_{oe,i}\right)^{\frac{n+1}{n}}\left[\left(\frac{r_{2e,i}^2-\left(r_{oe,i}\right)^2}{r_{2e,i}^2}\right)^{\frac{n+1}{n}}{}_2F_1\left(\frac{n+1}{n},\frac{3n+1}{2n};\frac{2n+1}{n};\frac{r_{2e,i}^2-\left(r_{oe,i}\right)^2}{r_{2e,i}^2}\right)\right] \tag{8}$$

Since the unknown $r_{oe,i}$ in the equation appears both inside the hypergeometric function and in its coefficients, and given that the hypergeometric function itself has no analytic solution, directly solving for $r_{oe,i}$ through algebraic manipulation is exceptionally challenging. Therefore, this paper presents a numerical method for solving the equation.

The original equation is reformulated as a root-finding problem: finding the root $r_{oe,i}$ that satisfies $F(r_{oe,i}) = 0$. Herein, the function $F(r_{oe,i})$ is defined as the difference between the two sides of Eq. (8), as expressed in Eq. (9):

$$F\left(r_{oe,i}\right)=\frac{nr_{oe,i}^{\frac{2}{n}}}{n-1}\left[\left(r_{oe,i}\right)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n},-\frac{1}{n};\frac{3n-1}{2n};1\right)-\left(r_1\right)^{\frac{n-1}{n}}{}_2F_1\left(\frac{n-1}{2n},-\frac{1}{n};\frac{3n-1}{2n};\frac{r_1^2}{r_{oe,i}^2}\right)\right]$$
$$-\frac{n}{2(n+1)}\left(r_{oe,i}\right)^{\frac{n+1}{n}}\left[\left(\frac{r_{2e,i}^2-\left(r_{oe,i}\right)^2}{r_{2e,i}^2}\right)^{\frac{n+1}{n}}{}_2F_1\left(\frac{n+1}{n},\frac{3n+1}{2n};\frac{2n+1}{n};\frac{r_{2e,i}^2-\left(r_{oe,i}\right)^2}{r_{2e,i}^2}\right)\right] \tag{9}$$

The numerical determination of the root is achieved through a combination of the following two techniques:

(1) Approximation of Hypergeometric Functions

The three Gaussian hypergeometric functions in Eq. (9) are approximated using a series expansion, as specified in Eq. (6). The summation is truncated at its first five terms to optimize for both computational accuracy and efficiency.

(2) Bisection Method for Root-Finding

Leveraging the Intermediate Value Theorem for continuous functions, the bisection method is employed within a pre-defined interval $[r_1, r_{2e,i}]$ that is confirmed to bracket the root. The iterative procedure is as follows:

① Calculate the midpoint *mid* of the current interval $[a, b]$ as $mid = a + \frac{b-a}{2}$.
② Evaluate the value of $F(mid)$ using Eq. (9), which involves calling the hypergeometric function approximation.
③ Check the sign of $F(mid)$: if $F(a)\cdot F(mid) < 0$,the root lies in the subinterval $[a, mid]$, update $b = mid$; otherwise, the root lies in $[mid, b]$,update $a = mid$.
④ Terminate the iteration and return mid as the root if either of the following criteria is met: the absolute function value satisfies $|F(mid)| < \epsilon$ (where ϵ is a preset tolerance, e.g., $\epsilon = 10^{-6}$), or the interval width $[a, b]$ is sufficiently small, i.e.,$\frac{b-a}{2} < \epsilon$.

5. Dimensionless volumetric flow rate in an eccentric annulus

Under concentric annular flow conditions, the relationship between the frictional pressure drop gradient G_{pf} and the dimensionless volume flow rate q_{aD} for power-law is given by [23]:

$$G_{pf}=\frac{k}{r_2}\left(\frac{Q}{r_2^3 q_{aD}}\right)^n \tag{10a}$$

$$q_{aD}=\frac{n\pi}{1+3n}\left(\frac{1}{2}\right)^{1/n}\left\{\left[1-\left(\frac{r_o}{r_2}\right)^2\right]^{1+1/n}-\left(\frac{r_1}{r_2}\right)^{1-1/n}\left[\left(\frac{r_o}{r_2}\right)^2-\left(\frac{r_1}{r_2}\right)^2\right]^{1+1/n}\right\} \tag{10b}$$

Similarly, the correlation between the eccentric frictional pressure drop gradient G_{pfe} and the eccentric dimensionless volume flow rate is expressed as:

$$G_{pfe}=\frac{k}{r_2}\left(\frac{Q_e}{r_2^3 q_{aDe}}\right)^n \tag{11a}$$

Upon substituting r_{oe} for r_o and r_{2e} for r_2 in Eq. (10b), and taking into account the symmetry of the eccentric annulus, the following is derived:

$$q_{aDe}=\frac{n}{1+3n}\left(\frac{1}{2}\right)^{1/n}\int_0^{\pi}\left(\frac{r_{2e}}{r_2}\right)^{3+1/n}\left\{\left[1-\left(\frac{r_{oe}}{r_{2e}}\right)^2\right]^{1+1/n}-\left(\frac{r_1}{r_{2e}}\right)^{1-1/n}\left[\left(\frac{r_{oe}}{r_{2e}}\right)^2-\left(\frac{r_1}{r_{2e}}\right)^2\right]^{1+1/n}\right\}d\theta \tag{11b}$$

Let I denote the integral term in Eq. (11b), where $f(\theta)$ is the integral kernel function. Thus:

$$I=\int_0^{\pi} f(\theta)d\theta$$

$$f(\theta)=\left(\frac{r_{2e}}{r_2}\right)^{3+1/n}\left\{\left[1-\left(\frac{r_{oe}}{r_{2e}}\right)^2\right]^{1+1/n}-\left(\frac{r_1}{r_{2e}}\right)^{1-1/n}\left[\left(\frac{r_{oe}}{r_{2e}}\right)^2-\left(\frac{r_1}{r_{2e}}\right)^2\right]^{1+1/n}\right\}$$

By equally partitioning the circumferential domain [0, π] of the eccentric annulus into m elements and approximating the flow within each subinterval $[\theta_{i-1},\theta_i]$ as concentric flow, where r_{oe} is approximated by $r_{oe,i}$, r_{2e} by $r_{2e,i}$, and $f(\theta)$ as a constant $f(\theta_i)$, we obtain:

$$I=\frac{\pi}{m}\sum_{i=1}^{m}f(\theta_i) \tag{12a}$$

$$f(\theta_i)=\left(\frac{r_{2e,i}}{r_2}\right)^{3+1/n}\left\{\left[1-\left(\frac{r_{oe,i}}{r_{2e,i}}\right)^2\right]^{1+1/n}-\left(\frac{r_1}{r_{2e,i}}\right)^{1-1/n}\left[\left(\frac{r_{oe,i}}{r_{2e,i}}\right)^2-\left(\frac{r_1}{r_{2e,i}}\right)^2\right]^{1+1/n}\right\} \tag{12b}$$

Substituting Eq. (12a) into Eq. (11b) yields:

$$q_{aDe}=\frac{n}{1+3n}\left(\frac{1}{2}\right)^{1/n}\frac{\pi}{m}\sum_{i=1}^{m}f(\theta_i) \tag{13}$$

6. Determination of hydraulic parameters in eccentric annuli

The analysis of hydraulic parameters in eccentric annuli, particularly under engineering data conditions, primarily focuses on the radial distributions of flow velocity, shear stress, and shear rate across distinct circumferential angular orientations, as well as the relationship between

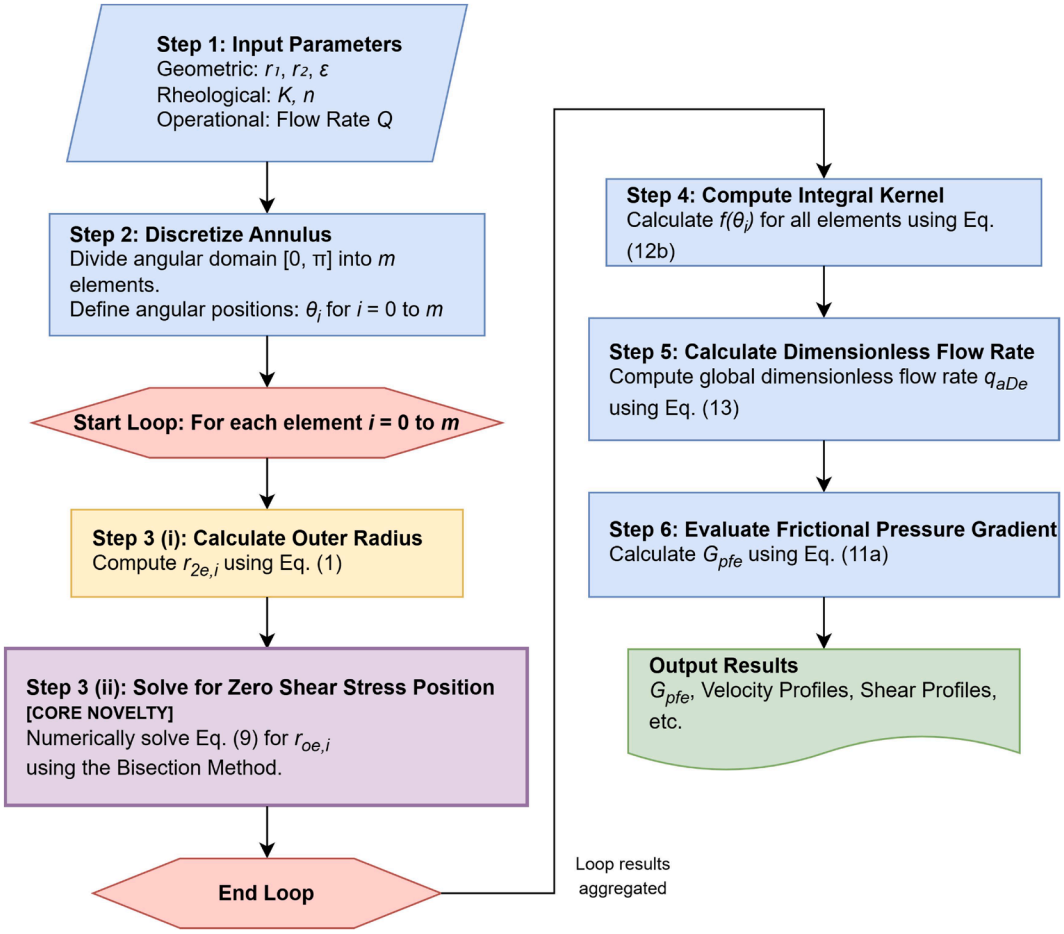


Fig. 2. Flowchart of the computational procedure for the proposed semi-analytical model.

Table 1
Zero shear stress positions and velocities.

θ/rad	Present method				Approximated method			
	r_{oe}/m	v_{oe}^{-}/m·s^{-1}	v_{oe}^{+}/ m·s^{-1}	Δv_{oe}/ m·s^{-1}	r_{oe}/m	v_{oe}^{-}/ m·s^{-1}	v_{oe}^{+}/ m·s^{-1}	Δv_{oe}/ m·s^{-1}
0	0.1337	2.8261	2.8257	0.0003	0.1345	3.0658	2.8886	0.177226
π/6	0.1301	2.4205	2.4212	0.0007	0.1309	2.6219	2.4768	0.145071
π/3	0.1207	1.5384	1.5386	0.0003	0.1213	1.6583	1.5780	0.080371
π/2	0.1092	0.7611	0.7607	0.0005	0.1095	0.8150	0.7832	0.031753
2π/3	0.0990	0.3305	0.3307	0.0002	0.0992	0.3521	0.3417	0.010394
5π/6	0.0926	0.1588	0.1594	0.0006	0.0927	0.1688	0.1649	0.003844
π	0.0904	0.1170	0.1175	0.0005	0.0905	0.1242	0.1217	0.002529

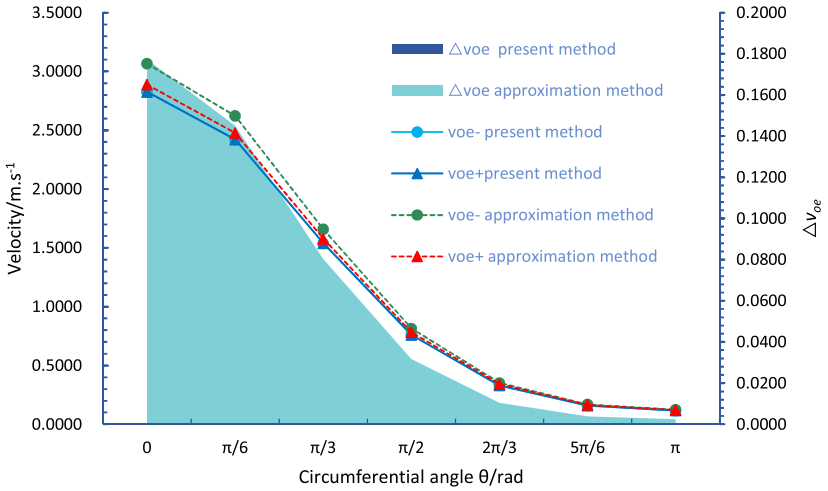


Fig. 3. Comparison of velocities calculated at the zero-shear-stress interface.

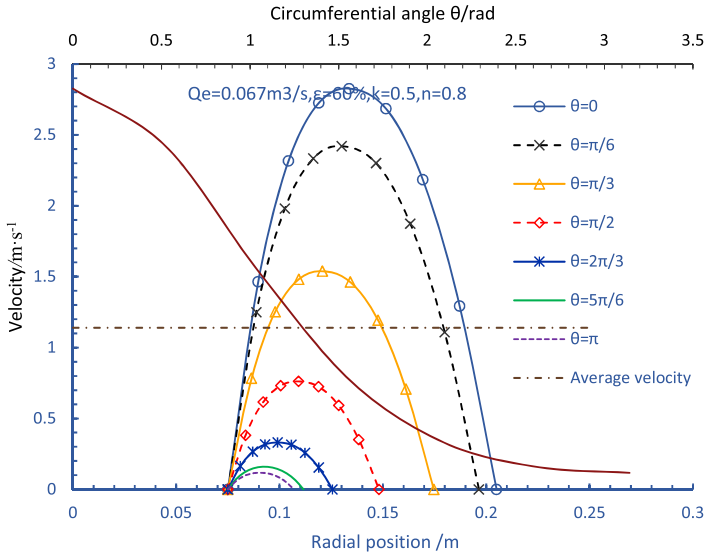


Fig. 4. Velocity profiles in eccentric annular flow.

the frictional pressure drop gradient and flow rate.

6.1. Shear stress and shear rate in eccentric annuli

By replacing the zero shear stress position r_o in the concentric annulus shear stress and shear rate expressions [23] with $r_{oe,i}$ within the sub element at circumferential angle θ_i in the eccentric annulus, we obtain:

$$\tau_{e,i} = \frac{Gp_{fe}}{2}\left[r_e - \frac{(r_{oe,i})^2}{r_e}\right] \tag{14}$$

$$\gamma_{e,i} = \pm\left(\frac{Gp_{fe}}{2k}\right)^{1/n}\left|r_e - \frac{(r_{oe,i})^2}{r_e}\right|^{1/n} \tag{15}$$

6.2. Frictional pressure drop gradient calculation for eccentric annuli

The computational procedure for determining the frictional pressure drop gradient and other hydraulic parameters is summarized in the step-by-step algorithm below. A schematic flowchart illustrating this entire process is provided in Fig. 2.

(i) Define geometric parameters (r_1, r_2, e), fluid rheological properties (K, n), and the volumetric flow rate Q.
(ii) Based on the geometric symmetry of the eccentric annulus flow, the angular interval $[0, \pi]$ is discretized into m equally spaced elements, with angular positions defined by $\theta_i = \frac{i\pi}{m}$ for $i = 0,\ldots,m$.
(iii) Compute the outer boundary radial position $r_{2e,i}$ for all elements using Eq. (1).
(iv) Apply the bisection method (as described in Section 4) to determine the zero shear stress position $r_{oe,i}$ for all elements.
(v) Compute $f(\theta_i)$ using Eq. (12b) for all elements.
(vi) Calculate the dimensionless volumetric flow rate q_{aDe} via Eq. (13).
(vii) Evaluate the frictional pressure drop gradient G_{pfe} using Eq. (11a).

7. Case study

This study investigates the radial profiles of velocity, shear stress, and shear rate as a function of circumferential angle θ, and the quantitative correlation between frictional pressure drop gradient and flow rate, for a power-law fluid in an eccentric annulus conFig.d by a 5-7/8" drill pipe and a 12-1/4" wellbore. Case parameters include: inner radius

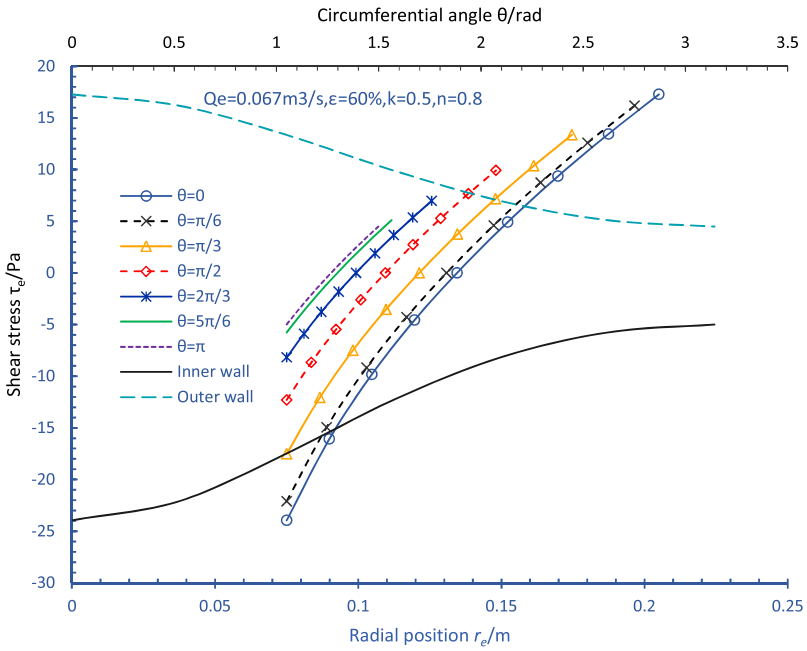


Fig. 5. Shear stress profiles in eccentric annular flow.

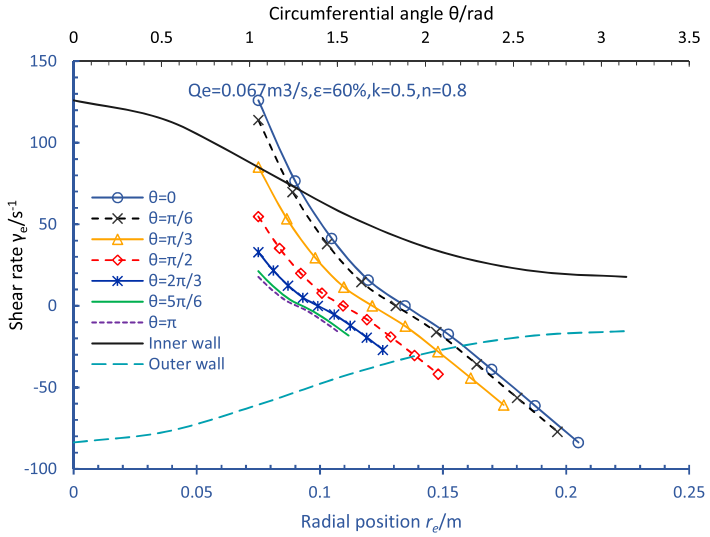


Fig. 6. Shear rate profiles in eccentric annular flow.

r_1=0.075m, outer radius r_2=0.156m, drill pipe standoff e=0.049m (eccentricity $\varepsilon = e/(r_2 - r_1)$=60 %),flow rate Q_e=0.067m^3/s,drilling fluid consistency index k=0.500, and flow behavior index n=0.800.

7.1. Velocity profile

In this study, seven representative circumferential angles (θ=0,π/6,π/3,π/2,2π/3,5π/6, and π) are selected within the eccentric annulus. For each of these angular directions, the zero shear stress positions r_{oe} are determined by applying both the present and approximation methods. Based on these calculated r_{oe} values, Equations(4a) and (4b) are then applied to determine the velocities v_{oe}^-, v_{oe}^+, and $\Delta v_{oe} = |v_{oe}^- - v_{oe}^+|$. The computational results are summarized in Table 1 and Fig. 3. Here, the 'Present method' refers to the numerical solution of Eq. (8) using the bisection method to find the precise $r_{oe,i}$. The 'Approximated method' refers to the use of the simplified, explicit formula given in Eq. (7) to estimate $r_{oe,i}$, a technique used in earlier models to avoid numerical complexity [21].

As evidenced in Table 1 and Fig. 3, significant discrepancies exist between the two methods in calculating the zero shear stress position r_{oe} and its corresponding subdomain velocities (v_{oe}^- and v_{oe}^+). According to the physical definition of the zero-shear-stress position, the velocities in adjacent subdomains must satisfy continuity ($v_{oe}^- = v_{oe}^+$) at this critical location. Therefore, the Δv_{oe} value directly indicates the accuracy in determining the zero shear stress point position.

Across all seven circumferential angular directions, the calculated velocity interface difference (Δv_{oe}) by the present method is consistently very small, with a maximum difference of only 0.0007 m/s. This indicates that the present method accurately determines the position of the zero-shear-stress point, resulting in nearly identical velocities calculated on both sides of this point. In contrast, the Δv_{oe} values computed by the approximation method are significantly larger, reaching as high as 0.1772 m/s and 0.1451 m/s at θ=0 and θ=π/6, respectively. These values are substantially greater than the corre-

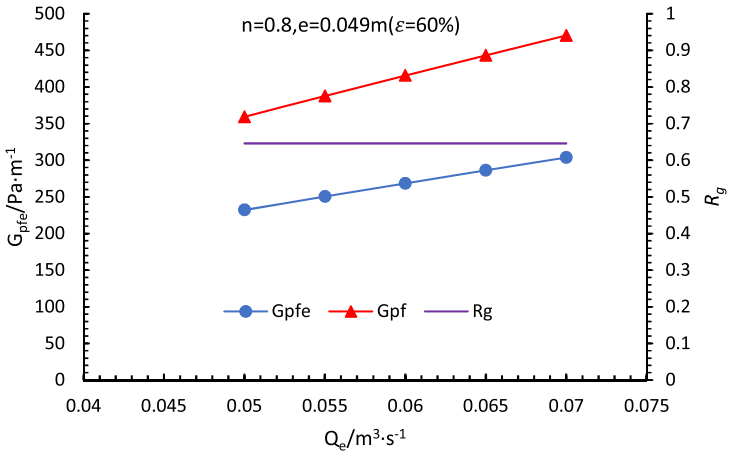


Fig. 7. Variations of frictional pressure drop gradients with flow rate.

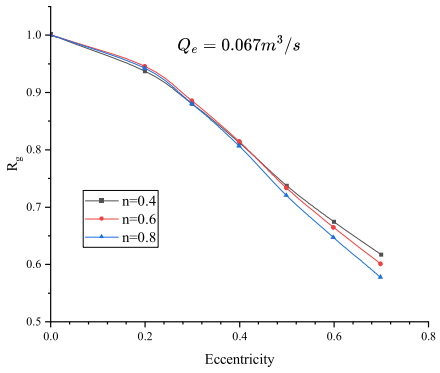


Fig. 8. Variations of R_g with eccentricity and flow index.

sponding differences obtained by the present method. This demonstrates that the approximation method introduces significant deviations in determining the position of the zero shear stress point, failing to ensure proper velocity matching between the sub-domains at that point.

The frictional pressure drop gradient calculated using the present method is 293.284 Pa/m, whereas the result obtained from the approximation method is 304.358 Pa/m. Since the present method more accurately satisfies the physical conditions locally (at the zero shear stress point), its calculated overall flow parameters also more accurately represent the actual conditions.

Applying the r_{oe} determined by the present method, the radial velocity distributions at seven different circumferential angles were calculated, as depicted in Fig. 4. The Fig.'s data indicate that the velocity distribution of the power-law fluid in the eccentric annulus exhibits high non-uniformity in the circumferential direction. As the circumferential angle θ increases from the wide-gap side ($\theta = 0$) to the narrow-gap side ($\theta = \pi$), the peak velocity shows a non-linear decreasing trend (from

Table 2
Comparison of predicted pressure gradients with Salubi data [12] for a low shear-thinning fluid (PL-1, $n = 0.75$).

Case No.	Eccentricity	G_{pf}(Pa/m)		Relative Error (%)
		Salubi [12]	Present Model	
S-1-A-1	0.00	154.29	156.04	1.13
S-1-A-3	0.50	112.97	114.64	1.47
S-1-A-5	0.70	93.856	92.57	1.37
S-1-B-1	0.00	587.931	598.24	1.75
S-1-B-3	0.50	448.006	439.52	1.9
S-1-B-5	0.70	357.498	354.91	0.72

Table 3
Comparison of predicted pressure gradients with Salubi data [12] for a highly shear-thinning fluid (PL-2, $n = 0.27$).

Case No.	Eccentricity	G_{pf}(Pa/m)		Relative Error (%)
		Salubi [12]	Present Model	
S-2-A-1	0.00	169.41	172.13	1.61
S-2-A-3	0.50	124.89	127.86	2.38
S-2-A-5	0.70	68.92	108.68	57.69
S-2-B-1	0.00	236.51	241.41	2.07
S-2-B-3	0.50	173.95	179.32	3.09
S-2-B-5	0.70	157.76	152.41	3.39

2.826 m/s at $\theta = 0$ to 0.117 m/s at $\theta = \pi$). Furthermore, the flow demonstrates significant zonal characteristics: peak velocities in the high-energy region ($\theta \leq \pi/3$) are notably higher than the average velocity (1.140 m/s), while those in the low-energy region ($\pi/3 < \theta < \pi$) are considerably lower than the average velocity ($v_{oe}/v_a \leq 0.667$).

7.2. Shear stress profile

Fig. 5 illustrates the radial distribution of shear stress at seven selected circumferential angles within the eccentric annulus, along with the variation of shear stress on the inner and outer walls as a function of circumferential angle. This Fig. indicates that the shear stress distribution within the eccentric annulus is highly non-uniform, simultaneously dependent on both radial position (r_e) and circumferential angle (θ). Shear stress exhibits a non-linear radial distribution, with extreme values occurring in the near-wall regions. The absolute shear stress at the inner wall is consistently higher than at the outer wall. Overall, the

Table 4
Comparison of predicted pressure gradients with experimental data from Ahmed and Miska [25].

Q (m^3/s)	G_{pf}(Pa/m)				
	Measured	Present Model (ε=0)	Relative Error (%, ε=0)	Present Model (ε=0.6)	Relative Error (%, ε=0.6)
0.000035	1764	2717.77	54.07 %	1840.02	4.31 %
0.000126	4018	5583.01	38.95 %	3779.89	5.93 %
0.000379	7644	10368.83	35.65 %	7020.06	8.16 %
0.000629	9898	13768.04	39.10 %	9321.45	5.82 %
0.000884	11270	16661.86	47.84 %	11280.67	0.09 %
0.001136	13622	19171.71	40.74 %	12979.93	4.71 %
0.001283	14014	20523.60	46.45 %	13895.20	0.85 %
AAPE (%)	-	-	43.26 %	-	4.27 %

absolute shear stress significantly decreases with increasing circumferential angle. Its maximum values are observed at the widest annulus gap ($\theta = 0$), reaching 23.949 Pa on the inner wall and 17.276 Pa on the outer wall. Conversely, the minimum values are located at the narrowest gap ($\theta = \pi$), being 4.994 Pa on the inner wall and 4.482 Pa on the outer wall. This pronounced circumferential disparity in shear stress is a core physical manifestation of the eccentric effect on the flow field structure and wall friction.

7.3. Shear rate profile

The radial shear rate profiles across seven selected circumferential angular directions within the eccentric annulus, along with the variations of shear rates at inner and outer walls as functions of angular coordinate θ, are shown in Fig. 6. It indicates that, the shear rate exhibits a nonlinear decreasing trend with increasing radial position at any given circumferential angle As the circumferential angle θ increases from the wide-gap side ($\theta = 0$) to the narrow-gap side ($\theta = \pi$), the shear rate at the inner wall monotonically decreases from 126.005 s^{-1} to 17.154 s^{-1}, while the shear rate at the outer wall monotonically increases from -83.772 s^{-1} to -15.509 s^{-1}.

7.4. Sensitivity analysis of frictional pressure drop gradient in eccentric annuli

To evaluate the model's response to key operational and physical parameters, a sensitivity analysis was performed. This analysis examines the influence of flow rate (Q_e), eccentricity (ε), and the flow behavior index (n) on the frictional pressure drop gradient. The ratio of the frictional pressure drop gradient in an eccentric annulus to that in a concentric annulus, R_g, is defined to quantify the eccentricity effect:

$$R_g = \frac{G_{pfe}}{G_{pf}} = \left(\frac{q_{aD}}{q_{aDe}}\right)^n \tag{16}$$

As both q_{aD} and q_{aDe} depend solely on the annulus geometrical configuration and the flow behavior index n, R_g is independent of the flow rate Q and the consistency index k.

(1) Influence of Flow Rate

Fig. 7 illustrates the relationship between the frictional pressure drop gradient (G_{pfe}) and the volumetric flow rate (Q_e) as Q_e increases from 0.05 m^3/s to 0.07 m^3/s, for a fixed eccentricity ($\varepsilon = 60$ %) and flow behavior index ($n = 0.8$). Given that n is close to Newtonian characteristics (n=1.0) in this case, the relationship exhibits an approximately linear growth, demonstrating the model's robust prediction of fundamental hydraulic trends.

(2) Influence of Eccentricity and Flow Behavior Index

Fig. 8 depicts the influence of eccentricity (ε) and the flow behavior index (n) on the pressure drop ratio (R_g). The data in the Fig. indicate that (R_g exhibits a significant negative correlation with eccentricity. As the drill pipe eccentricity ε increases from 0 to 0.9, the (R_g value decreases dramatically; for instance, at ε=0.7, it drops to values between 0.577 and 0.617, signifying a pressure drop reduction of up to 42.3 % compared to the concentric case. This phenomenon is primarily attributed to the fluid flowing preferentially through the wider, lower-resistance gap created by the increased eccentricity.

The effect of the drilling fluid flow behavior index n on R_g is also highly dependent on eccentricity. At low eccentricities ($\varepsilon \leq 0.4$), the influence of n is negligible, with the curves for n=0.4, 0.6, and 0.8 being nearly indistinguishable. However, at high eccentricity ratios ($\varepsilon > 0.4$), the curves corresponding to different flow behavior indices begin to diverge. In this regime, fluids with a higher n (closer to Newtonian) correspond to lower R_g values compared to fluids with a lower n (more shear-thinning). This quantitatively demonstrates a nuanced interaction where the pressure reduction effect of eccentricity is more pronounced for less shear-thinning fluids.

7.5. Model validation

The accuracy and reliability of the proposed semi-analytical model were rigorously validated against two distinct types of high-fidelity data from established literature: (i) recent, comprehensive numerical simulation data from Salubi et al. [12], and (ii) classic experimental data from Ahmed and Miska [25]. This two-pronged approach enables benchmarking against both highly controlled computational results and real-world experimental measurements, encompassing a diverse range of conditions.

(1)Validation against Numerical Simulation Data

Salubi et al. [12] performed detailed CFD simulations to investigate the combined effects of fluid rheology, inner pipe rotation, and eccentricity. Their published data provides an ideal benchmark for validating the present model due to its high precision and the systematic variation of parameters. For this validation, we selected data for two distinct Power-Law (PL) fluids: a low shear-thinning fluid (PL-1, n=0.75) and a highly shear-thinning fluid (PL-2, n=0.27). The validation covers concentric (ε=0) and eccentric (ε=0.5, ε=0.7) cases at various flow rates, without inner pipe rotation, to focus on the model's core capability in predicting pressure gradients under axial flow conditions.

The comparison between the pressure gradients predicted by the present model and the data from Salubi et al. [12] is presented in Tables 2 and 3. As shown in Table 2, for the low shear-thinning fluid (PL-1, n=0.75), the model demonstrates excellent agreement with the benchmark data, with all relative errors being less than 2.2 % and an AAPE of only 1.44 %. This confirms the model's high accuracy for fluids with moderate shear-thinning behavior.

For the highly shear-thinning fluid (PL-2, n=0.27), as detailed in Table 3, the model also performs robustly. Except for one data point, all relative errors remain below 3.6 %. A significant deviation of 57.69 % is observed for case S-2-A-5, which corresponds to a high eccentricity (ε=0.7) at a low flow rate. This data point is identified as an apparent anomaly within the benchmark data, as it contradicts the established physical principle that pressure gradients decrease with increasing eccentricity in laminar flow. Our model's prediction (108.68 Pa/m) is numerically different but physically consistent with the overall trend. Excluding this anomalous data point, the AAPE for the PL-2 fluid is an excellent 2.54 %.

(2)Validation against Experimental Data

To further test the model's performance against real-world conditions, a validation was performed using the experimental data from Ahmed and Miska [25]. The experiments were conducted on a Power-Law fluid (n=0.56, K=0.833 Pa·s^n) in a 1.0-in. × 1.5-in. annulus (k = 0.67) without pipe rotation. The original authors noted that their experimental setup had an inherent eccentricity, resulting in measured pressure losses that were significantly lower than the theoretical concentric predictions. This scenario provides a perfect test case for the capabilities of our eccentric model.

Table 4 presents a comprehensive comparison of the experimental measurements against our model's predictions for both concentric (ε=0) and eccentric (ε=0.6) configurations. The concentric model predictions consistently overestimate the measured pressure gradient, with a large AAPE of 43.3 %. This confirms that an idealized concentric model is inadequate for describing the experimental reality.

However, when applying the proposed semi-analytical model with a physically reasonable eccentricity of ε=0.6, the predictions align remarkably well with the experimental data. The AAPE plummets to just

4.3 %, demonstrating the model's profound accuracy in capturing the pressure-reducing effect of eccentricity. This excellent agreement transforms a theoretical discrepancy into a strong validation of the model's physical fidelity and its practical utility in analyzing real-world, non-ideal annular geometries.

8. Discussion, limitations, and future work

The results demonstrate that the proposed semi-analytical model is a powerful tool for hydraulic analysis, offering a significant advantage in accuracy over previous approximate superposition methods and a computational efficiency advantage over full numerical simulations. The detailed velocity and shear profiles (Figs. 4-6) it generates are critical for advanced drilling engineering analyses, a capability that empirical correlations lack.

However, it is essential to acknowledge the limitations of the current study, which also highlight avenues for future research.

(i) Flow Regime: The model is derived for steady-state, laminar flow and does not apply to transitional or turbulent regimes.
(ii) Fluid Rheology: The current framework is formulated for power-law fluids. While fundamental, this is a simplification, as many drilling fluids exhibit yield stress. Advanced numerical studies by Azouz et al. [8], Hashemian et al.[13], and Dokhani et al. [9] have successfully modeled the more comprehensive Yield-Power-Law (YPL) fluids. Extending the present semi-analytical approach to incorporate yield stress effects would be a valuable area for future work.
(iii) Operational and Geometric Assumptions: The model assumes a stationary inner pipe and does not account for rotation. Furthermore, the model's performance at extreme eccentricities (e.g., $\varepsilon > 0.95$) has not been exhaustively tested and may require further investigation.

9. Conclusions

(1) A numerical method for determining the zero shear stress point, based on hypergeometric function approximation and the bisection method, significantly improved the precision of its radial position calculation. This precision is a key contribution of this work; comparative validation shows that common approximation methods can lead to significant velocity matching errors at the zero shear stress point (up to 0.0007 m/s in the case study), which directly affects the accuracy of hydraulic parameter calculations.
(2) A semi-analytical-numerical hybrid method was formulated for directly calculating shear stress, shear rate, velocity profiles, and frictional pressure drop gradient within eccentric annuli, based on given geometric parameters, rheological parameters, and volumetric flow rate. This provides a comprehensive hydraulic characterization tool that overcomes the limitation of previous models, which could not directly solve for pressure drop from a given flow rate.
(3) The model quantitatively captures the pronounced non-uniformity and asymmetry in radial and circumferential directions. As the circumferential angle θ increases from the wide-gap side ($\theta = 0$) to the narrow-gap side ($\theta = \pi$), the peak velocity, absolute shear stress at the inner/outer walls, and absolute shear rate at the inner/outer walls all significantly and monotonically decrease within the eccentric annulus. Concurrently, the absolute shear stress and shear rate at the inner wall are higher than at the outer wall. These detailed profiles, unavailable from simple correlations, are critical for analyses such as hole cleaning and torque-and-drag modeling.
(4) The model confirms that eccentricity is a critical factor influencing the frictional pressure drop in eccentric annuli. As eccentricity increases, the ratio of frictional pressure drop gradient in eccentric to concentric annuli ($\boldsymbol{R_g}$) significantly decreases. When the eccentricity reaches 0.7, this ratio drops to values between 0.577 and 0.617, representing a reduction in pressure drop of up to 42 %.
(5) Under high eccentricity conditions ($\varepsilon > 0.4$), the influence of the drilling fluid's flow behavior index n on the frictional pressure drop ratio becomes more pronounced. Specifically, the model predicts that higher n fluids (closer to Newtonian) exhibit lower frictional pressure drop gradients than lower nn fluids (more shear-thinning), an unusual effect that is important for fluid design in highly deviated wells

Nomenclature

$(a)_k$: Pochhammer symbol, rising factorial
γ: Shear rate, s^{-1}
$\gamma_{e,i}$: Shear rate in the θ_i direction of the eccentric annulus, s^{-1}
ϵ: Eccentricity ratio, $\epsilon = e/(r_2 - r_1)$, dimensionless
θ: Circumferential angle, *rad*
θ_i: Circumferential angle of the i-th sub-region in the eccentric annulus, *rad*
τ: Shear stress, Pa
$\tau_{e,i}$: Shear stress in the θ_i direction of the eccentric annulus, Pa
a: Hypergeometric function parameter, dimensionless
b: Hypergeometric function parameter, dimensionless
B: Incomplete Beta function, dimensionless
c: Hypergeometric function parameter, dimensionless
e: Drill string standoff, m
$f(\theta)$: Dimensionless integral kernel function for eccentric annulus volumetric flow rate, dimensionless
${}_2F_1$: Gaussian hypergeometric function, dimensionless
G_{pf}: Frictional pressure drop gradient in concentric annulus, Pa/m
G_{pfe}: Frictional pressure drop gradient in eccentric annulus, Pa/m
h: Eccentric annulus gap width, m
I: Integral term
k: Power-law fluid consistency index, Pa·s^n
m: Number of sub-regions in eccentric annulus, dimensionless
n: Power-law fluid flow behavior index, dimensionless
q_{aD}: Dimensionless volumetric flow rate in concentric annulus dimensionless
q_{aDe}: Dimensionless volumetric flow rate in eccentric annulus, dimensionless
Q_e: Volumetric flow rate in eccentric annulus, m^3
r_1: Inner radius of annulus, m
r_2: Outer radius of annulus, m
$r_{2e,i}$: Outer wall radial position of eccentric annulus in θ_i direction, m
r_e: Radial position in eccentric annulus, m
R_g: Ratio of frictional pressure drop gradient in eccentric to concentric annulus, dimensionless
$v_{e,i}$: Drilling fluid velocity in the θ_i direction of the eccentric annulus, m/s
z: Hypergeometric function independent variable, dimensionless

CRediT authorship contribution statement

Xuncheng Song: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

I declare that I have no financial, business, or other competing interests with any company or entity that could be affected by the research reported in the enclosed paper.

Data availability

Data will be made available on request.

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